Huawei BTS-3900 Access Node Hardware/Environmental Fault Resolution SOP (BTM-ACC-34)

Node: BTM-ACC-34 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_1109

Description

This Standard Operating Procedure (SOP) addresses the failure domain of hardware/environmental faults on the Huawei BTS-3900 access node, specifically Node ID BTM-ACC-34, located at BTM Layout Aggregation. The failure is triggered by ASIC memory parity errors, line card crashes, or power supply redundancy loss, resulting in the ALARM_1109 alarm with a minor severity. The underlying cause is attributed to the Huawei BTS-3900's role as an access node, which relies on the Advanced Switching Interconnect (ASI) architecture. The ASI architecture uses a combination of ASICs and line cards to process and switch traffic. In the event of a hardware or environmental failure, the ASI architecture's redundancy mechanisms are triggered, attempting to switch traffic to redundant line cards or ASICs. However, if the failure is severe enough to impact the power supply or if the redundant components are also affected, the node may fail to recover, resulting in the ALARM_1109 alarm. This SOP outlines the diagnostic and resolution steps to address this failure domain and restore normal node operation.

Pre-requisite Checks

- Step: Verify Physical Site Conditions, Description: Ensure the site is accessible and physically secure, with all necessary power and environmental monitoring equipment functioning correctly.

- Step: Check Redundant Link Status, Description: Verify the status of the redundant links and line cards to determine if any have failed or are unavailable.

- Step: Gather Historical Data, Description: Review historical logs and alarm records to identify any previous similar failures or issues with the node.

Potential Root Causes

Cause	Probability
ASIC Memory Parity Errors	High
Line Card Crashes	Medium
Power Supply Redundancy Loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display current-configuration | include asic, Expected Output: ASIC memory parity errors: 0, Description: This command checks for ASIC memory parity errors by displaying the current configuration
```

```
and including the ASIC section. The expected output indicates that
there are no ASIC memory parity errors.

Command: display linecard status, Expected Output: Linecard status:
Normal, Description: This command checks the status of the line cards
by displaying the line card status. The expected output indicates that
the line cards are functioning normally.

Command: display power-supply status, Expected Output: Power supply
status: Normal, Description: This command checks the status of the
power supplies by displaying the power supply status. The expected
output indicates that the power supplies are functioning normally.
```

Resolution Steps

Step 1: Step: Replace Faulty ASIC or Line Card, Command: replace linecard , Description: This step involves replacing the faulty ASIC or line card by issuing the replace linecard command. The command replaces the faulty component with a new one, which should resolve the issue.

Step 2: Step: Reboot the Node, Command: system reboot, Description: This step involves rebooting the node to ensure that the replacement component is recognized and the node is fully operational. The system reboot command restarts the node, which should resolve the issue.

Step 3: Step: Verify Node Operation, Command: display node status, Description: This step involves verifying the node's operation by displaying the node status. The display node status command checks the node's overall status, including the status of the replacement component.

Post-Check Verification

```
Command: display node status, Expected Output: Node status: Normal,
Description: This command verifies the node's operation by displaying
the node status. The expected output indicates that the node is
functioning normally.

Command: display linecard status, Expected Output: Linecard status:
Normal, Description: This command verifies the status of the line
cards by displaying the line card status. The expected output
indicates that the line cards are functioning normally.

Command: display power-supply status, Expected Output: Power supply
status: Normal, Description: This command verifies the status of the
power supplies by displaying the power supply status. The expected
output indicates that the power supplies are functioning normally.
```

Roll-back Procedures

- Step: Revert Changes, Command: undo replace linecard , Description: This step involves reverting the changes made during the resolution process by issuing the undo replace linecard command. The command reverts the replacement component to its previous state.

- Step: Reboot the Node, Command: system reboot, Description: This step involves rebooting the node to ensure that the reversion process is complete. The system reboot command restarts the node, which should resolve the issue.